

# Cortical EEG slowing in resting-state Parkinson's disease

Final presentation: model, code, and findings

PD EEG group · Rui, Jan, Melissa, Friedrich · 26.06.2026

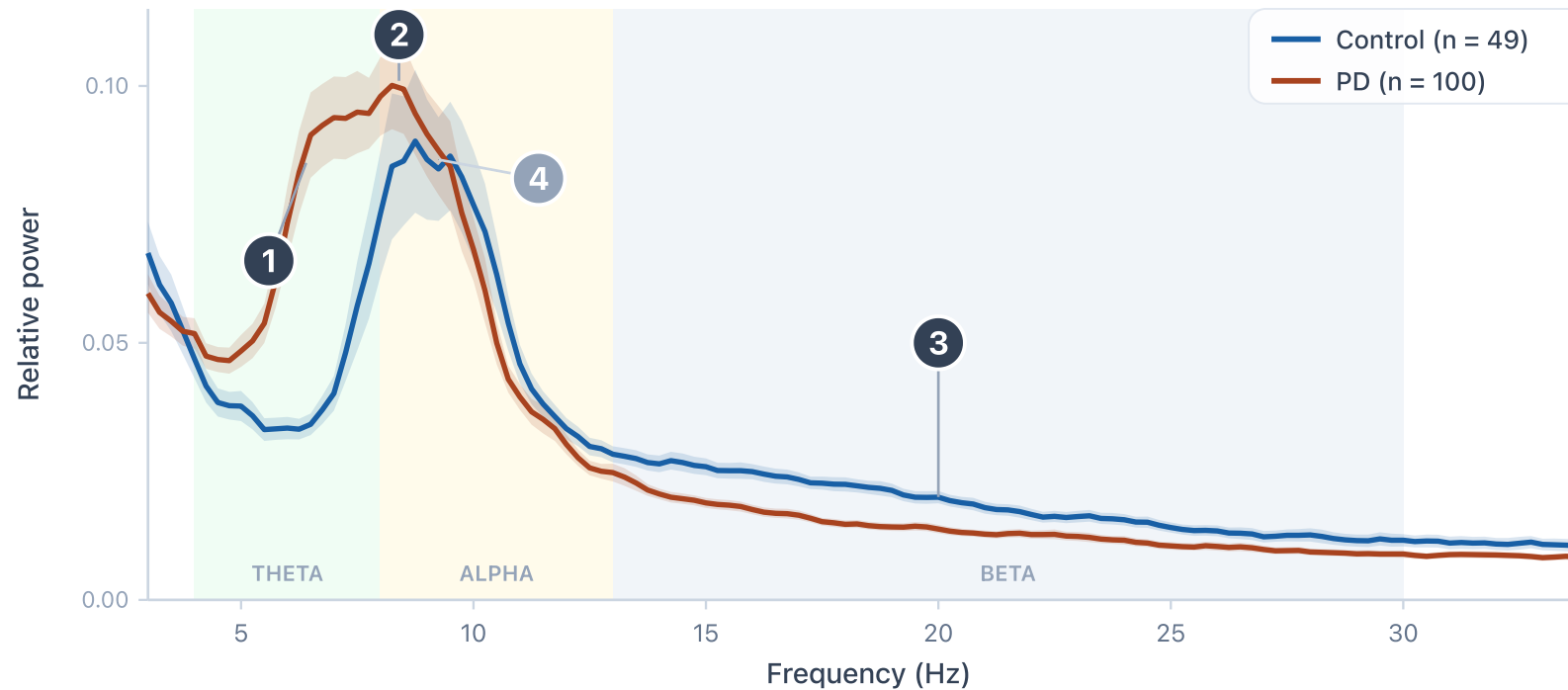
# Hypotheses

① theta power higher  
in PD ·  $p < 0.001$

② alpha peak slows  
9.25 → 8.00 Hz ·  $p < 0.001$

③ beta power lower  
in PD ·  $p < 0.001$

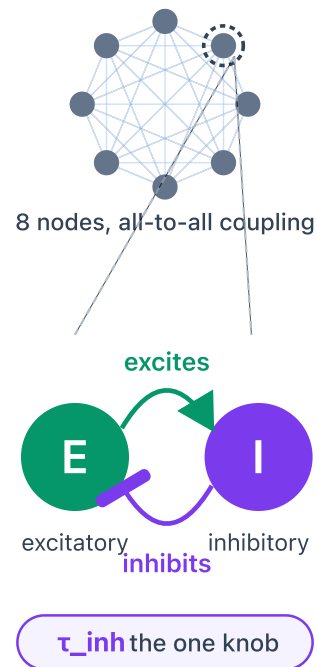
④ alpha power  
unchanged ·  $p = 0.9$



Three changes confirmed, one prediction null: the slowing is in frequency, not amplitude.

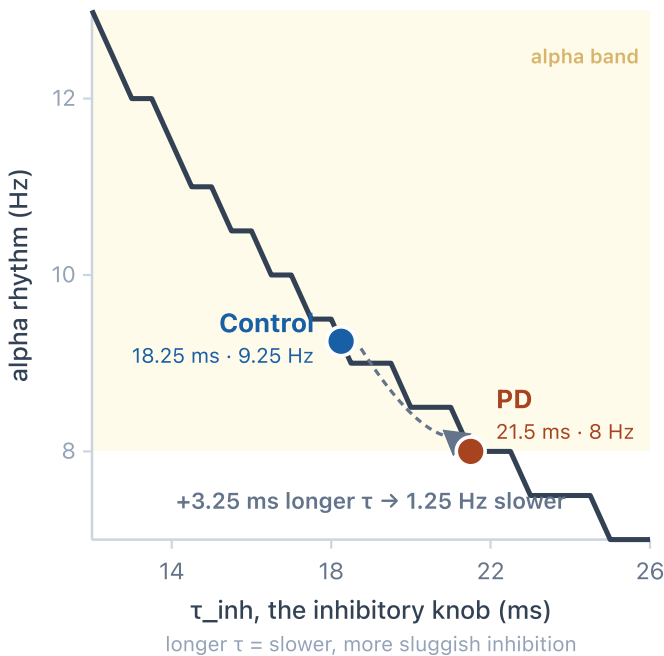
# The model: a Wilson-Cowan network

## THE MODEL: AN E-I LOOP

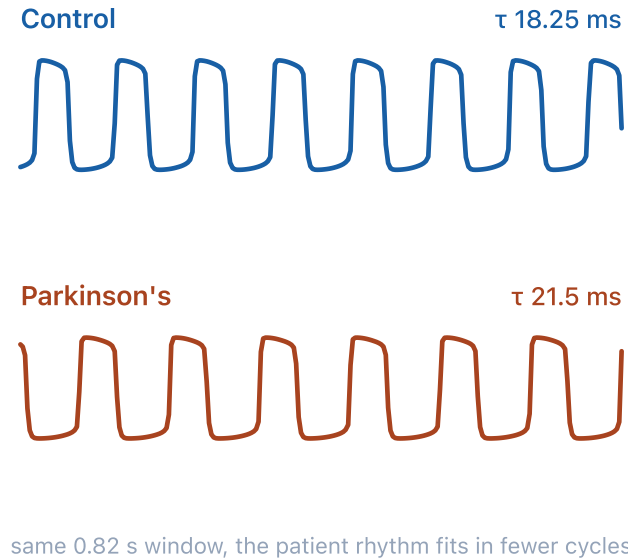


## TURN $\tau_{inh}$ UP, THE RHYTHM SLOWS

each point = the model run at one knob setting



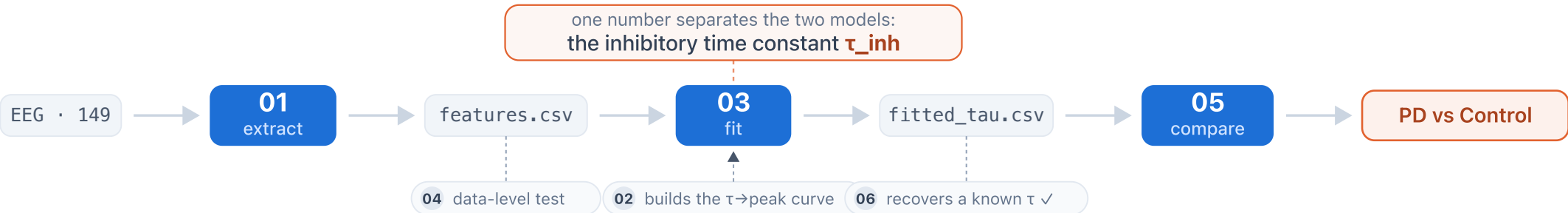
## THE SIMULATED RHYTHM



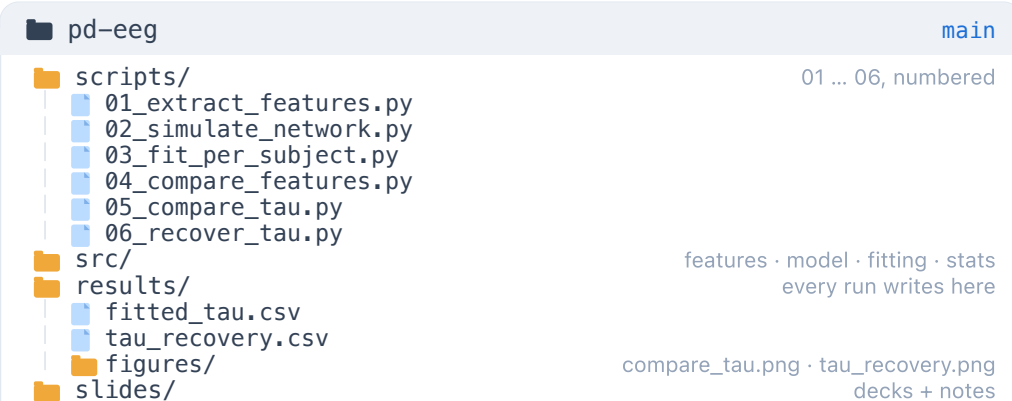
The whole change from healthy to patient is one number: a longer inhibitory time constant  $\tau_{inh}$ , about **+3.25 ms** in PD.

# Code & repository

THE PIPELINE EEG to result, through six numbered scripts



THE REPOSITORY

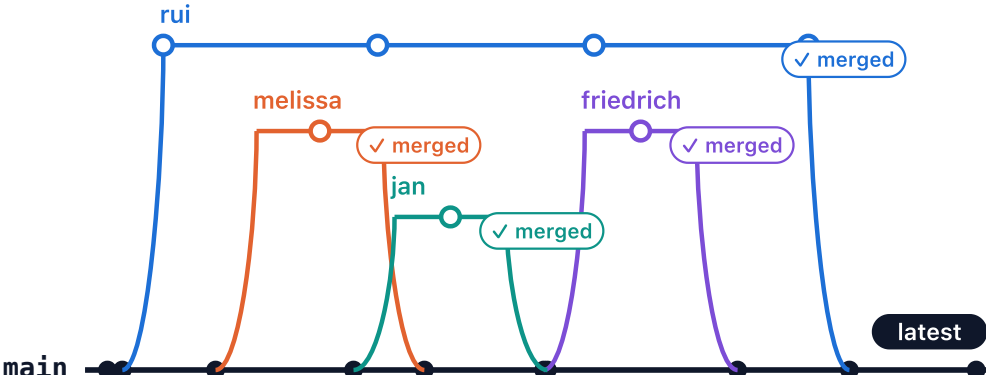


every result is one script run

```
bash
$ python scripts/05_compare_tau.py
Control 18.25 → PD 21.50 ms diff +3.25 p = 0.00025 ***
→ wrote results/figures/compare_tau.png
$ python scripts/06_recover_tau.py
22 known tau recovered bias -0.10 max err 1.75 ms
→ wrote results/tau_recovery.csv
```

VERSION CONTROL

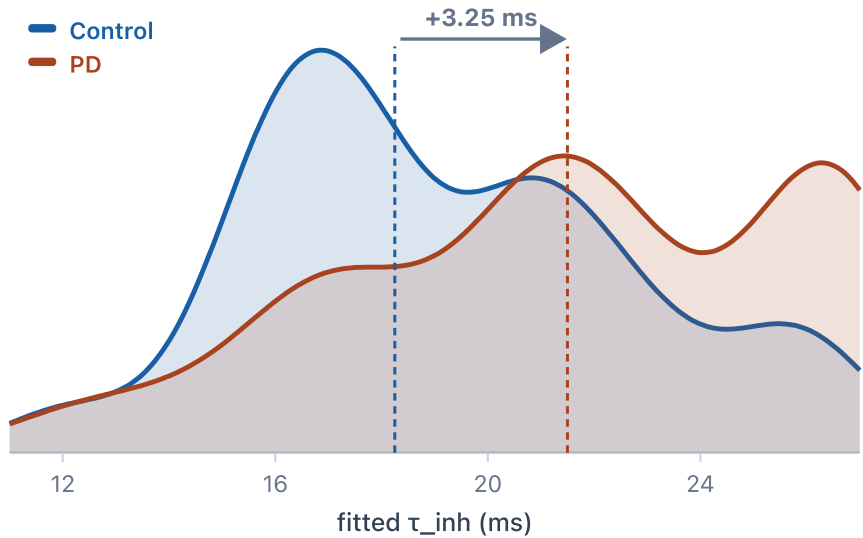
feature branches, reviewed, merged into main



50 commits · feature branches → reviewed → merged into main · fully reproducible

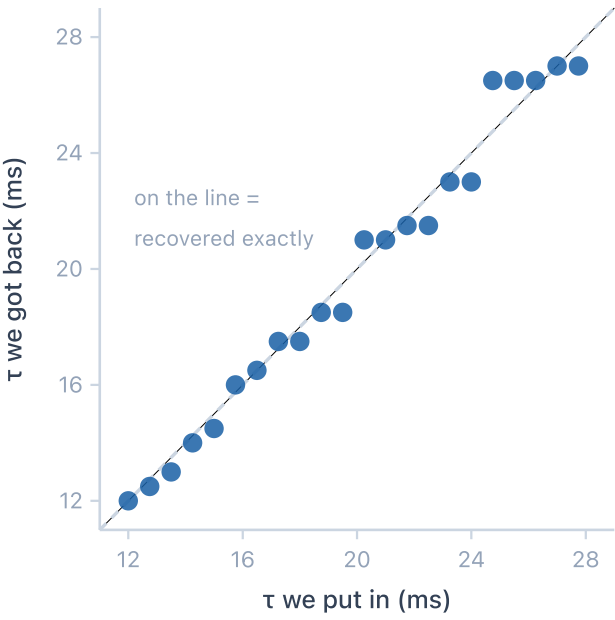
# Does the model separate patients from controls?

THE SEPARATION      fitted  $\tau_{inh}$ , Control vs PD



Control 18.25 vs PD 21.5 ms at the median.  
The distributions overlap: a random PD is higher  $\approx 68\%$  of the time.

PARAMETER RECOVERY      put a known  $\tau$  in, fit it back



**The fit is unbiased.**  
mean error  $-0.1$  ms  
typical error  $\pm 0.5$  ms

Fitted  $\tau_{inh}$  re-expresses the measured slowing as a model parameter (Mann-Whitney  $p \approx 0.0002$ ).

# Outlook and open challenges

---

## CURRENT CHALLENGES

- One parameter ( $\tau_{inh}$ ) reproduces the alpha slowing, but not the theta and beta changes
- The fit re-expresses the data-level effect, rather than adding independent evidence

## FUTURE DIRECTIONS

- Fit more than the peak (full spectrum or functional connectivity), on a real connectome
- Relate fitted  $\tau$  to clinical scores and to known inhibitory (GABAergic) changes in PD

# What we took away

---

**Rui** [your biggest learning]

**Jan** [your biggest learning]

**Melissa** [your biggest learning]

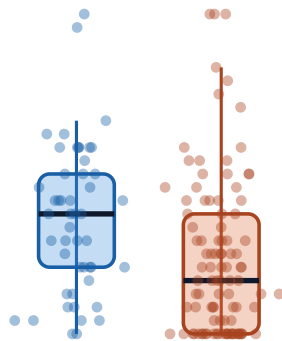
**Friedrich** [your biggest learning]

# Appendix: per-subject distributions

---

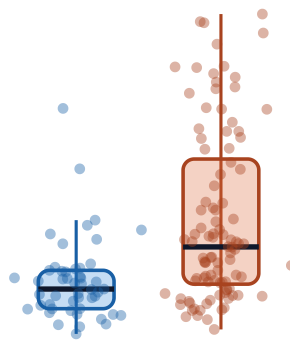
## PD vs Control: per-subject EEG features

Alpha peak frequency  
slower in PD



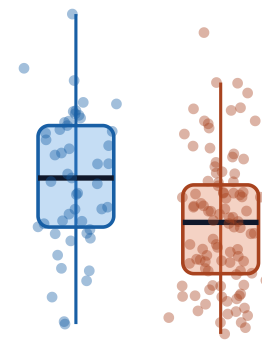
Control PD

Theta power  
higher in PD



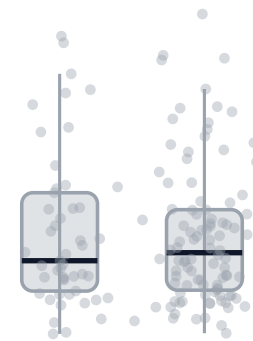
Control PD

Beta power  
lower in PD



Control PD

Alpha power  
unchanged



Control PD



# Appendix: effect sizes

## How Parkinson's changes the resting EEG

effect size per feature, with 95% confidence interval

